

Cover Letter — Submission to Journal of Earthquake and Tsunami

Manuscript title: A Saturation-Aware Multi-Stage Framework for Intensity-Driven Adaptive P-Wave Time Window Selection in Real-Time Spectral Acceleration Prediction: Operational Design for the Java-Sunda Subduction Zone

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To the Editor-in-Chief, Journal of Earthquake and Tsunami (JET), World Scientific Publishing Company

Dear Editor,

We are pleased to submit our manuscript “**A Saturation-Aware Multi-Stage Framework for Intensity-Driven Adaptive P-Wave Time Window Selection in Real-Time Spectral Acceleration Prediction: Operational Design for the Java-Sunda Subduction Zone**” for consideration as a regular research article in *Journal of Earthquake and Tsunami*.

Scientific contribution

This work introduces **IDA-PTW** (Intensity-Driven Adaptive P-wave Time Window), a saturation-aware multi-stage framework for real-time prediction of 5%-damped spectral acceleration $Sa(T)$ at 103 spectral periods. The pipeline operates in four cascaded stages: an Ultra-Rapid P-wave Discriminator (URPD) at $t = 0.5$ s, an XGBoost+LightGBM intensity gate at $t = 3$ s, a single-station distance ablation, and an adaptive PTW spectral regressor ensemble that elongates the observation window from 3 s to 5 s or 8 s based on predicted intensity class.

Trained on 338 events (M_w 1.7–6.2, R_{epi} 0.7–299.9 km) recorded at 151 IA-BMKG accelerograph stations along the Java-Sunda subduction zone between November 2022 and March 2026, the framework achieves the following headline metrics:

- Composite $R^2 = 0.7091$ across 103 spectral periods (operational $R^2 = 0.6962$)
- Stage-0 URPD AUC = 0.9136 at 0.5-s budget
- Stage-1 ensemble balanced accuracy = 81.68% with $\sim 86\%$ recall on the Damaging class
- End-to-end latency of 10.65 s, meeting the InaTEWS Golden-Time budget for 99.44% of traces
- $\Delta R^2 = +0.067$ over an Atkinson–Boore subduction GMPE baseline computed on the same data and CV protocol
- Al Atik (2010) sigma decomposition: $\sigma_{tot} = 0.862$ at $T = 1.0$ s (mean 0.755 across 103 periods)

Why JET

The manuscript is well aligned with JET’s scope on (i) earthquake source physics applied to operational early warning, (ii) ground-motion characterisation for subduction settings, and (iii) regional preparedness frameworks for tsunamigenic seismic zones. The Java-Sunda Trench, on which our dataset and pipeline are focused, is one of the principal source regions for Indian Ocean tsunamigenic earthquakes, and the framework

is designed for direct integration into the operational InaTEWS network operated by the Indonesian Agency for Meteorology, Climatology and Geophysics (BMKG).

The submission complements recent JET work on Indonesian EEWS context (Mukhlisin et al. 2023; Subekti et al. 2022; Pribadi et al. 2023; Irwansyah et al. 2024), state-of-the-art adaptive PTW review (Tatar et al. 2025), and tsunami warning for the Sunda Trench source zone (Srivastava et al. 2008).

Novelty

To the best of our knowledge, no prior study has presented a single deployable framework that simultaneously addresses all four operational gaps for tropical subduction EEWS: (i) saturation-awareness for $M_w > 6.5$ events, (ii) intensity-driven adaptive P-wave windowing on $\{3, 5, 8\}$ -s schedules, (iii) catalog-independent autonomous operation without external magnitude estimates, and (iv) full 103-period $Sa(T)$ prediction in real time. The methodological contributions include a hybrid 90-feature dictionary organised in four families with VIF below 10, a Bayesian-correct posterior-marginalisation formulation in the class-conditional regressor, and an honest negative result for single-station distance regression at Stage 1.5.

Reproducibility

The complete pipeline, trained models, evaluation metrics, and reproduction scripts are released at <https://github.com/hanif7108/IDA-PTW> (commit 3d06700). The repository is currently access-controlled during peer review for novelty protection; reviewers may obtain read access through the journal editor by request to the corresponding author. Upon manuscript acceptance, the repository will be made publicly available under the MIT licence together with a permanent Zenodo-archived release.

A bundled `verify_repro.py` script provides a single-command reproducibility audit (51 checks against the pinned environment, input checksums, and headline metrics). End-to-end FULL re-runs validate canonical metrics to within ± 0.001 of the reported values.

Compliance

- **Originality:** This manuscript has not been published elsewhere and is not under consideration by any other journal.
- **Authorship:** All listed authors have contributed substantively to the design, execution, analysis, or interpretation of the work, and have approved this submission.
- **Conflicts of interest:** The authors declare no competing interests.
- **Data ownership:** The IA-BMKG accelerograph waveforms used in this study are owned by the Indonesian Agency for Meteorology, Climatology and Geophysics (BMKG). Access for verification is available upon institutional request through the corresponding author.
- **AI use disclosure:** Large language model tools (Claude Code by Anthropic) were used for language editing, internal consistency auditing, and bibliographic verification. All research design, dataset curation, pipeline implementation, training, evaluation, statistical analysis, and substantive interpretation are the original work of the authors. Details are provided in the Acknowledgments section.

Suggested reviewers (optional)

The authors respectfully suggest the following reviewers, who have published in adjacent areas of EEWS, subduction-zone seismology, or machine-learning ground-motion prediction:

1. **Prof. Carmine Galasso**, University College London (UK) — EEWS algorithms, decision-making frameworks

2. **Dr. Mehrdad Tatar**, IIEES (Iran) / GFZ Potsdam — Adaptive PTW review, subduction EEWS
3. **Prof. Tzu-Yi Lin**, National Taiwan University — P-alert, large earthquake EEWS
4. **Prof. Aditya Riadi Gusman**, GNS Science (NZ) / Indonesian-affiliated — Tsunami source inversion, Indonesia

(The above is a courtesy suggestion list; the editor's discretion is, of course, paramount.)

Length and structure

- Manuscript length: $\approx$ 12,000 words excluding references and figure captions
- Number of figures: 16 (PDF/PNG, high resolution)
- Number of tables: 19
- References: 108 entries

The manuscript follows the World Scientific JET strict template (10 pt Times Roman, 6×9 in page geometry, single column, 13 pt baseline line spacing) and is provided in both PDF and LaTeX source form together with the supplemental materials (SM1–SM7).

Closing

We believe this work contributes a deployment-ready, scientifically rigorous, and regionally grounded earthquake early warning framework that advances the state of the art on adaptive P-wave windowing and saturation-aware spectral prediction for tropical subduction zones. We are grateful for your consideration and look forward to feedback from the editorial team and reviewers.

Sincerely,

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On behalf of: Dede Djuhana · Adhi Harmoko Saputro · Sigit Pramono

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